Nirbhay Beri

Engineer — Automation & Digitalisation | M.Tech (Smart Manufacturing), IISc Bangalore

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Summary

Professional in Smart Manufacturing and Industrial Digitalisation, with a growing focus on **AI**, **IIoT**, **digital-twin**, and **IT–OT** convergence. Brings a systems-oriented approach to understanding processes, connecting operational and information layers, and contributing to modern, future-ready manufacturing environments. Works closely with engineering, operations, and R&D teams to support clear decision-making, improve workflow clarity, strengthen industrial reliability, and contribute to smoother execution across digital and operational pipelines.

Professional Experience

Engineer — Automation & Digitalisation — Jul 2024 – Present

Atlas Copco Group (GECIA), Pune

- Designing an AI-assisted vision system for inspection of reworked **vacuum pumps**; configured the NVIDIA **Jetson Xavier**, established the data and model pipeline, and planned communication to PLCs for automated **pass/fail** decisions. Future roadmap includes integration with a **Fanuc** robot and **AMR** for automated parking to next station or rework area.
- Spearheading the end-to-end development of the **GECIA Industry 4.0 Lab**; responsible for **designing** the technical infrastructure, defining future-use **hardware specifications**, and managing **vendor liaisons** for full facility setup.
- Implemented **MES–PLC communication** for the AC New MES (.NET) using **Kepware** drivers via **OPC UA**, enabling reliable WorkStart/Stop and Andon events; supported successful go-live across two **conveyor lines** at the PNE Plant.
- Developed process simulations in **Visual Components** (Python API) for BMW, GM, and multiple Customer Centers, accelerating layout validation and reducing planning time.
- Designed concepts for **vision-based quality inspection** (O-Ring and Kit-Fill), integrating barcode capture, PLC handshake, and SQL logging to strengthen traceability and inspection consistency.
- Delivered the **LinguaLink translation** POC by fine-tuning language models on technical corpora, improving documentation creation efficiency.
- Built an IIoT telemetry pipeline using **Azure IoT Hub** to ingest environmental data (temperature/humidity) from **Advantech WISE 4250** modules via MQTT; visualized real-time metrics in Power BI for operational monitoring.
- Developed Ignition diagnostic dashboards for MES–PLC event monitoring at the AC PC PNE operations.
- Prototyped **AR-based** product visualisation experiences using **Unity**, **WebSockets**, and **HoloLens 2** for interactive demonstrations.
- Developing an Azure pipeline to ingest operator performance data from systems such as **Azumuta** and **BPCS** for efficiency analysis; building Power BI dashboards to visualise assembly output, quality trends, and operational KPIs. Future scope includes **predictive analytics** and **digital-twin** integration.

Research Intern — AI & Industrial Metaverse — May 2023 – Jul 2023

Siemens Technology, Bengaluru

- Developed a **virtual factory** environment in Unity, visualized via **HTC Vive headset**, to enable immersive evaluation and validation of production layouts.
- Built and validated a machine-layout optimisation framework using **evolutionary methods**, improving placement decisions based on throughput, ergonomics, and safety; results published in a Springer conference paper.

Education

M.Tech — Smart Manufacturing (With Distinction) **2022–2024**
Indian Institute of Science (IISc), Bengaluru
CGPA: 8.7/10

B.Tech (Honors) **2018–2022**
Dr. A.P.J. Abdul Kalam Technical University (AKTU), Lucknow
CGPA: 9.0/10

Key Projects

Multimodal XR System for Remote Human–Robot Welding & Assembly MTech Thesis

- Developed a unified XR framework combining visual, auditory, text and haptic feedback to enhance remote robot operation for assembly and welding tasks.
- Built a **VR-based welding** module using a digital twin (URDF) synchronized with a **DOBOT robot** via custom API; added multimodal interfaces including haptics, voice commands, and GelSight-inspired tactile sensing.
- Implemented **APF-based haptic guidance** for collision avoidance, automatic weld-path generation from CAD, and conducted user validation for trajectory accuracy using Discrete Fréchet Distance.
- ***Tech****: Unity (VR/MR), C#, Haptics (APF), HoloLens 2, HTC Vive, Digital Twin (URDF), GelSight, Python (Flask), Computer Vision, Robot Teleoperation.*

VR & ML for Maintenance Learning (Industrial Metaverse) IISc / 2023

- Developed a VR-based maintenance training environment for assembly/disassembly of industrial equipment.
- Integrated a **1D-CNN** fault-detection model (CWRU dataset) to visualize real-time ML predictions inside VR.
- ***Tech****: VR, Unity, 1D-CNN, TensorFlow, Data Visualization, Python, CAD modeling.*

AI & Vision-Based Robot for Surface Defect Detection IISc / 2022

- Developed a robotic inspection system using a DOBOT Magician with a **calibrated camera** to automate detection of cracks, pitting and fretting on **3D-printed aircraft** components.
- Implemented robot path planning (**lawnmower scan pattern**) for systematic coverage and built an image acquisition pipeline with accurate camera calibration.
- Finetuned model on custom dataset (Roboflow-augmented) to classify four defect types, demonstrating feasibility of automated NDE for quality control.
- ***Tech****: YOLOv8, OpenCV, DOBOT Robotics, Path Planning, Camera Calibration, Roboflow, NDE, Python.*

Publications

- **HaM3D: Generalized XR-Based Multimodal HRI Framework**
  Journal on Multimodal User Interfaces (Springer), 2024.
- **Enhancing Production Efficiency Using GA-Assisted Visualization**
  I-4AM Conference (Springer), 2023.

Skills

Automation & Industry 4.0
IT/OT Convergence, OPC UA, MQTT, Basic PLC Programming (TIA Portal, SIMATIC Manager, ISP Soft), Ignition, Visual Components (Simulation), Kepware, OPC Router.

AI, Robotics & Computer Vision
DOBOT Magician, FANUC Robots, Machine Learning (TensorFlow, Scikit-Learn), Computer Vision, NLP, GenAI, Data Analysis (Pandas, NumPy, Matplotlib).

Cloud, Software & Platforms
Microsoft Azure (IoT Hub, Logic Apps, Functions, Stream Analytics, ML), Unity 3D, Node-RED, Power Platform, SQL Server, MongoDB, InfluxDB, Grafana.

Hardware & Edge Computing
HoloLens 2, HTC Vive, Jetson Xavier/Nano, IoT Modules (WISE 4250), Industrial Cameras, Haptic Devices (3D Systems Touch).

Programming & Methodologies
Python, C#, GitHub, Cross-functional Collaboration, Innovative Problem Solving.

Key Academic Coursework
Digital Manufacturing, Sensors & Transducers, HMI, Operations Management, Additive Manufacturing, Creative Engineering Design, Probability & Statistics,Computing for AI & ML, NLP, Computer Vision, Data Modelling.

Achievements & Certifications

- **Star of the Month (Aug 2025)** — Atlas Copco Group.
- **Co-organized Industrial Robotics Workshop (KARMA)** — VIT Pune.
- **GATE 2022** — All India Rank 26.
- **B.Tech Honors** — Awarded Honors Degree (1 of 3 students).
- **Siemens Technology Research Fellowship** — 2022–2024.
- **NPTEL (Industrial IoT 4.0)** — Top rank.